Data Science In Chemical Engineering

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Contents

Chapter 1

Introduction

This course is designed to meet the evolving demands in the chemical industry due to the rising importance of data science. It aims to impart the fundamental skills required for understanding and leveraging the data science pipeline, including data collection, data management, knowledge extraction, and application of optimization techniques.

Throughout the program, methodologies for data retrieval, network graphs, and adherence to the FAIR principles (Findability, Accessibility, Interoperability, and Reusability) for data management will be explored.

Practical applications of data mining techniques, machine learning, statistical analysis, and model prediction for real-time analysis and decision-making using big data will be faced. Through hands-on practical sessions and real-world case studies, students will experience direct applications of these concepts in chemical engineering.
By the end of the course, students are expected to:

- Demonstrate a deep understanding of the fundamental concepts of the data science pipeline, including data collection, management, and knowledge extraction. They also understand and apply the principles of data management in accordance with the FAIR principles.

- Apply advanced data mining, machine learning, and deep learning techniques to effectively utilize big data. They demonstrate practical skills in implementing common algorithms for clustering, regression, and classification, including Neural Networks and Deep Neural Networks.

- Develop reduced-order models for real-time analysis, model prediction, and model quality assessment. They evaluate and critically assess the quality and effectiveness of data-driven decision-making models.

- Communicate complex data science concepts and results clearly and effectively to both specialist and non-specialist audiences. They demonstrate proficiency in presenting methodologies and findings in written and oral forms.

- Develop advanced learning skills that enable continuous self-improvement in data science and its applications in chemical engineering. They stay updated with the latest methodologies and adapt to new tools and techniques in the field.

The course will cover the following topics:

- **Data science and its mathematical foundations**: Overview of data science, its applications in engineering, and an introduction to machine learning. Foundational elements of calculus and linear algebra will be recalled as they underpin several machine learning algorithms.

- **Common optimization techniques**: Optimization is introduced as a core concept in data science, with topics covered including an introduction to optimization techniques, problems admitting closed-form solutions, and the exploration of convex optimization.
- **Data collection, crawling, and management**: Strategies and tools for massive data collection, handling missing data, principles of data storage and management, quality evaluation, and automatic conversion from data lakes to data warehouse solutions. The FAIR principles of data management will also be introduced.
- **Exploratory data analysis**: Basic statistical analysis, correlation, data visualization, and feature engineering.
- **Data retrieval**: Strategies for efficient data retrieval, with a particular focus on the use of network graphs.
- **Knowledge extraction and machine learning**: Application of techniques such as neural networks, classification, regression, and clustering in the chemical industry. The goal is to leverage the potential inherent in large datasets for process optimization and control.
- **Advanced machine learning and reduced-order models**: In-depth exploration of advanced machine learning techniques, including supervised and unsupervised learning methods. Development and application of reduced-order models for real-time analysis of big data. This includes model prediction, functional estimation for model quality assessment, and consequent decision making.
- **Deep learning**: Introduction to deep learning techniques, including setting up neural networks, understanding activation functions, and training techniques such as backpropagation.

One or more seminars led by industry experts will be included, providing real-world context to the application of data science in various sectors of chemical engineering, and potential avenues for future careers in the field.

The practical sessions will provide hands-on experience in Python for data cleaning, visualization, model building, and interpretation.

Chapter 2

Python Installation

2.1 Anaconda

To use Python efficiently and in an organized manner, we will set up an Anaconda environment to ease the lives of everyone here.

See here for the installation guide `https://www.anaconda.com/docs/getting-started/anaconda/install`

2.2 Visual Studio Code

For interacting with Python code and other tools, it is highly recommended to use VS Code as your development environment.

See here for the installation: `https://code.visualstudio.com/download`

2.3 Github

All the codes and the material related to the practical sessions will be shared using the appropriate WeBeep pages and (mainly) the GitHub repository of the course.

2.4 Detailed steps

Once you have installed VS Code and Anaconda, you are ready to begin the setup of the Python environment (i.e., Python + all the necessary additional packages).

1. Clone (i.e., have a copy on your pc) the GitHub repository of the course. Go to `https://github.com/Riccaraccio/Data-Science`. It is recommended to use Git to manage this step. Alternatively, you can download it as a compressed folder. You can follow the step-by-step guide of **How to clone a GitHub repository**.

2. On Windows, open the **Anaconda prompt**. On the other platforms, use directly the terminal where you installed Anaconda. Navigate to the folder where you have cloned the repository using "`cd /path/to/Data-science`".

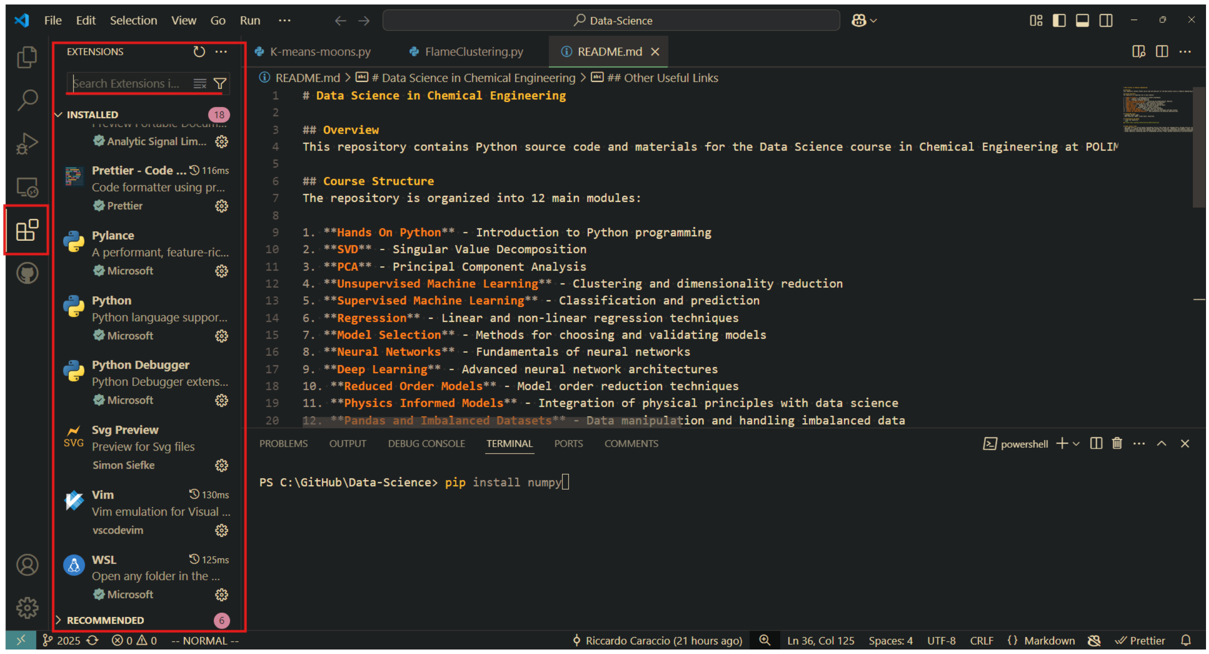
3. type "`conda create -n data-science python=3.12`". This creates the environment where all the necessary packages will be stored.

4. Now do "`conda activate data-science`". This activates the environment.

5. Lastly, "`pip install -e .`". Do not forget the last dot; this installs all the necessary packages.

2.5 Extensions

Visual Studio Code extensions are add-ons that enhance the functionality of your code editor. They're like apps you can install to make coding easier and more productive. You can install them from the corresponding tab, as shown below:



For a better coding experience, we suggest that you install the default **Python** extension, **Pylance**, **Python Debugger** and the **Jupiter** one. Many more are available in the environment; various let you personalize the appearance of the code (Search 'Theme' for this kind of stuff), and others add additional help to support you in writing code. You decide!

Chapter 3

Example

Let's try to run a simple example to verify the installation. In a folder of your preference, open a new Python Notebook file (es `filename.ipynb`) in VS Code.

See here for details on how the notebook work:`https://code.visualstudio.com/docs/datascience/jupyter-notebooks`

Verify that the Anaconda environment is correctly detected by VS Code selecting it as at kernel for the notebook. If not detected at first do `Select another kernel -> Python Environments -> Creat Python Enviroment -> Enter interpreter path`. Next navigate to your Anaconda installation folder (some thing like: `/path/to/Anaconda/envs/data-science`) and selecting the python.exe in the correct folder.

In the new source file, paste the following code:

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 10, 100)
y = np.sin(x) + np.random.normal(0, 0.2, 100)
plt.plot(x, y, label='Random Data')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('Numpy and Matplotlib Example')
plt.legend()
plt.show()
```

Figure 3.1 shows an example of how this would look. Run the code using the button at the top.

Try to understand what this code is doing, it's very intuitive.

If everything goes smoothly you have successfully completed the instillation procedure and you are ready to follow the course.

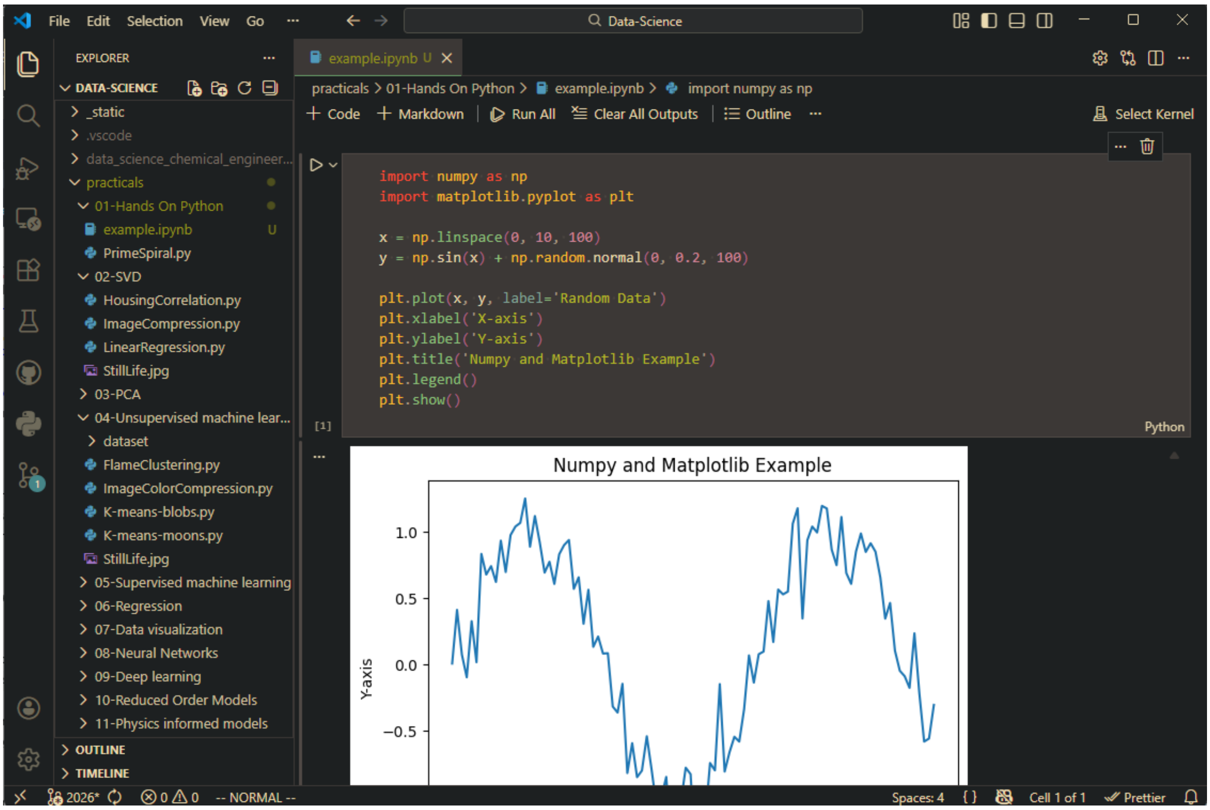


Figure 3.1: Exaple Notebook